Oxidation states

Learning outcomes

By the end of this section you should be able to:

- Deduce the oxidation states of an atom or an ion in a compound.
- Explain why the oxidation state of an element is zero.

Metallic and non-metallic character is the tendencies of atoms to lose or gain electrons. Can you think of any ways that we can track how many electrons are transferred or distributed when a chemical bond is formed? How did we keep track of this when we looked at ionic bonding in section S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/)? When we think of a substance like NaCl it is easy to identify that sodium has transferred an electron to chlorine forming Na^+ and Cl^- ions.

But what about in a substance like H_2O? There is no electron transfer occurring between oxygen and hydrogen, but the two atoms also do not share electrons equally in their polar covalent bond. These electrons are unevenly distributed. So how might we track this electron distribution of atoms in any substance, regardless of bonding type?

In ionic compounds, we tracked how electrons were transferred by considering the ionic charges of the positive and negative ions. Ionic charges however are only specific to ionic compounds. A similar but more inclusive electron tracking method involves assigning an oxidation state or oxidation number to each atom in a substance. Oxidation state is the numerical, hypothetical charge assigned to an atom in a substance based on the degree of electron transfer between atoms. These values are based on the relative electronegativities of elements.

Study skills

Oxidation state and oxidation number are two commonly interchangeable terms with the same meaning.

You might be wondering how is an oxidation state different from an ionic charge? Oxidation states are hypothetical and a method to track how electrons are distributed. They may be used in substances where electron transfer has not occurred at all. Ionic charges on the other hand indicate a numerical quantity of electrons that has been either lost or gained from a species. To differentiate the two terms a slightly different notation is used. **Table 1** shows the different notations used for the ion Na^+. Note that since sodium has lost an electron in Na^+ the numerical value is the same for both the oxidation state and ionic charge.

Table 1. Comparing ionic charge and oxidation state of Na^+.

Na^+	
Ionic charge	**Oxidation state**
1+	+1

Rules for assigning oxidation states

In order to assign the oxidation states for substances, the following rules can be applied. For each rule we will look at examples involving application of the rule.

1. A pure element will have an oxidation state of 0. This is because the distribution of electrons between atoms is the same for atoms of the same element. Atoms of the same element will have the same tendency to attract or gain electrons, therefore an oxidation number of zero is used to indicate no degree of electron transfer.

Element	Oxidation State
Na(s)	Oxidation state of Na: 0
Cl_2(g)	Oxidation state of each Cl: 0
P_4(s)	Oxidation state of each P: 0
C(s)	Oxidation state of C: 0

2. Monoatomic ions have an oxidation state equal to the charge of the ion.

Element	Oxidation State
Cu^{2+}	Oxidation state of Cu: +2

Element	Oxidation State
Cl^{-}	Oxidation state of each Cl: −1
S^{2-}	Oxidation state of each S: −2
Fe^{3+}	Oxidation state of Fe: +3

3. Oxidation state of all atoms in a neutral compound (ionic, molecular or both) must sum up to give zero overall.

4. Oxidation states of atoms in polyatomic ions must add up to give the overall charge of the ion.

OH^{-}	
O	Oxidation number on O: −2
H	Oxidation number on H: +1
Sum	−2 + 1 = −1
To sum we would add the oxidation number of the oxygen atom and the hydrogen ator	

5. Oxidation numbers of atoms are typically the charge of the most common ion. The typical oxidation numbers for the following groups are:

- Group 1 metals: +1 oxidation number.
- Group 2 metals: +2 oxidation number.
- Group 17 elements: –1 oxidation number (unless bonded to another more electronegative halogen).
- Oxygen: –2 oxidation number (unless in a peroxide, O_2^{2-}, then it is –1)
- Hydrogen: +1 oxidation number (unless in a metallic hydride such as NaH).

LiCl		MgF_2	
Li	Oxidation number on Li: +1	Mg	Oxidation number on Mg: +2
Cl	Oxidation number on Cl: −1	F	Oxidation number on each F: −1

LiCl		MgF_2	
Sum	+1 + (−1) = 0	Sum	+2 + 2(−1) = 0
Compounds are neutral so the sum of the oxidation states must be equal to zero.			

6. Assign positive oxidation numbers to the most metallic elements first and negative oxidation numbers to the most non–metallic elements first. Since metallic character is defined as the tendency of an atom to lose electrons and non–metallic character as the tendency of an atom to gain electrons these are the elements that will have the most predictable oxidation state. In most cases, this involves working from the elements on the outside of the chemical formula and working inwards.

$KMnO_4$		Na_3PO_4	
K is the most metallic element and O is the most non–metallic element.		Na is the most metallic element and O i most non–metallic element.	
K	Oxidation number on K: +1	Na	Oxidation number on Na: +1
O	Oxidation number on each O: −2	O	Oxidation number on each O:
Sum	+1 + *x* + 4(−2) = 0	Sum	3(+1) + *x* + 4(−2) = 0
Can use the variable *x* as a placeholder to solve for the oxidation state of the last element. the overall substance is neutral the oxidation states must sum to zero.			
Mn	Oxidation number on Mn: +7	P	Oxidation number on P: +5

Study skills

When a substance is made up of more than two elements it is helpful to set up a mathematical equation in order to algebraically solve for the oxidation number of the element of interest.

Worked examples

Try assigning oxidation states following the three worked examples shown below.

Worked example 1

Deduce and compare the oxidation states of each element in NO and NO_2.

Both NO and NO_2 are made up of only nitrogen and oxygen. Since both elements are non–metals, we can assign the oxidation number to oxygen first since it has a higher non–metallic character and a greater tendency to gain electrons.

NO			
O	Oxidation state of O: −2	O	Oxidation st
Sum	−2 + x = 0	Sum	2(−2) + x =
Can use the variable x as a placeholder to solve for the oxidation st overall substance is neutral the oxidation states must			
N	+2	N	+4

Therefore the oxidation states of nitrogen and oxygen are +2 and −2 in NO and +4 and −2 in NO_2.

In both substances, the oxidation states of nitrogen are positive. In NO_2 the oxidation state of nitrogen is a greater positive number since it is bonded to electronegative oxygen atoms

Worked example 2

Assign oxidation numbers to each element in $Al_2(SO_4)_3$.

Looking at each of the elements in $Al_2(SO_4)_3$, aluminium is the most metallic element and oxygen is the most non–metallic element. This means we will assign the oxidation numbers to Al and O first and then solve for S last.

Since Al is in group 13 and has 3 valence electrons its oxidation number would be +3.

Since O is in group 16 and has 6 valence electrons, its oxidation number would be −2.

We can now create a mathematical equation summing the oxidation numbers using the variable x as a placeholder for S. The sum of all the oxidation numbers will equal 0 since $Al_2(SO_4)_3$ is an overall neutral substance. Be sure to account for the subscripts in your mathematical equation noting that there are 2 Al atoms, 3 S atoms and 12 O atoms.

$$2(+3) + 3(x) + 12(-2) = 0$$

Rearranging and solving for *x* we find:

$$x = \frac{+24 - 6}{3}$$
$$x = +6$$

Therefore the oxidation numbers of each element in $Al_2(SO_4)_3$ are:

- Al: +3
- S: +6
- O: –2

Below is a tricky worked example. Try answering it yourself first and then read through the solution.

Worked example 3

Assign oxidation numbers to each element in hydrogen peroxide, H_2O_2.

This is a tricky example. And the reason it is tricky is that you will get a different answer depending on which element you assign the oxidation number to first. Let's look at both possible options and reason through which answer makes more sense.

H_2O_2			
Assigning H first		Assi	
H	+1	O	−2
Sum	2(+1) + 2*x* = 0	Sum	2*x* + 2(−2
O	−1	H	+2

You will notice that if we assign hydrogen first oxygen will have an oxidation number of –1 and if we assign oxygen first hydrogen will have an oxidation number of +2. Can you identify an issue with one of these answers?

The oxidation number of +2 does not make sense for hydrogen. This is because hydrogen only has one electron. It is not possible for hydrogen to have an oxidation number of +2 because there are not two electrons to lose.

Therefore the oxidation numbers of each element in H_2O_2 are:

- H: +1
- O: –1

This is an example of a peroxide. The peroxide anion also has this same exception of oxidation number for oxygen. This can happen when metals are bonded to the peroxide anion, such as in the example of Na_2O_2. Oxygen in this compound will have the –1 oxidation state in comparison to Na_2O we are more familiar with where oxygen will have the –2 oxidation state.

Making connections

Oxidation numbers are very useful for analysis of redox reactions. By assigning oxidation numbers to each element in the reactant and products it is easy to analyse how electrons have been redistributed throughout a chemical reaction. Redox reactions are a type of electron transfer reactions that you will learn more about in section R3.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/).

Naming conventions using oxidation numbers

When naming substances containing elements that can have variable oxidation states, Roman numerals are used to indicate the oxidation state. You have already looked at examples that do this in section S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) when you looked at naming ionic compounds with cations made up of a transition element. For example in $CuCl_2$, Cu has the +2 oxidation state therefore the name of the ionic compound is copper(II) chloride.

Roman numerals can also be used to show the oxidation states of elements within oxyanions. Oxyanions are negatively charged polyatomic ions involving oxygen bonded with another element. When naming oxyanions, the oxidation state of the element bonded to oxygen is indicated using Roman numerals. For example in section S3.1.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-and-nonmetallic-continuum-id-44695/) when we looked at the formation of H_2SO_3 and H_2SO_4 acids from oxides of water dissolving in water, these acids can also be named sulfuric(IV) acid and sulfuric(VI) acid. The Roman numeral indicates the oxidation number on the sulfur since that is the element bonded to oxygen making up the anion part of the substance. Can you identify which name belongs to which acid?

Table 2 shows some common examples of substances involving transition metals and oxyanions that use Roman numerals to indicate oxidation states. The generic names are also included as these are names that do still persist in chemistry despite not being recognised as proper naming by IUPAC.

Table 2. Common examples of substances involving transition metals and oxyanions.

Name of compound	Defining oxidation number	IUPAC name (Roman numerals)	Generic nar
$FeCl_2$	Fe: +2	iron(II) chloride	ferrous chlor
$FeCl_3$	Fe: +3	iron(III) chloride	ferric chlori
$Na_2Cr_2O_7$	Cr: +6	sodium dichromate(VI)	sodium dichro
$KMnO_4$	Mn: +7	potassium manganate(VII)	potassium permangana
Na_2SO_4	S: +6	sodium sulfate(VI)	sodium sulfa
Na_2SO_3	S: +4	sodium sulfate(IV)	sodium sulf
$Ca(NO_3)_2$	N: +5	calcium nitrate(V)	calcium nitra
$Ca(NO_2)_2$	N: +3	calcium nitrate(III)	calcium nitr

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Communication skills – Using terminology, symbols and communication conventions consistently
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Oxidation number pixel art

Complete the following oxidation number pixel art (https://docs.google.com/spreadsheets/d/1jSJBUttypl-cGmtEixhzNYBNfB3ACNNEEnuAViFVZql/copy?) on Google sheets. You will need to make a copy of the document for yourself. There are 20 substances where you will need to correctly identify the oxidation number of the specified element. For each substance you identify correctly part of an image will be revealed. When all 20 oxidation numbers have been identified the entire pixel art image will be revealed.